

— SYSTEMS MANIFESTO - 2026

10 SYSTEMS. 10 PROBLEMS. ONE STUDIO.

A technical blueprint of the software systems MIKESTH3TIC.DEV is building to solve the most consequential problems facing Africa and the world — engineered from Nairobi, designed for scale.

DISRUPT · AUTOMATE · DOMINATE

10 SYSTEMS	20+ PRODUCTS SHIPPED	08 AI SYSTEMS BUILT	∞ SCALE CEILING
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10 systems. Each one built around a real problem that currently has no adequate software solution.

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// ENGINEERING PHILOSOPHY

Every system in this document is built around a specific, measurable problem. The architecture, feature set, and technology choices are driven entirely by what the problem requires — not by what is fashionable, not by what is easiest, and not by what looks impressive in a deck.

MIKESTH3TIC.DEV builds software that works in the real conditions of the markets it serves — low connectivity, constrained devices, multilingual users, and regulatory environments that paper-based systems were designed around. Each system is engineered to those constraints from day one, not retrofitted to them after launch.

Mobile Money Infrastructure

A real-time financial transaction platform for the unbanked — built for feature phones, 2G networks, and agent-based liquidity.

// PROBLEMS IT SOLVES

×	2.5 billion adults lack access to any formal financial service
×	Remittances move through informal, untracked, high-risk channels
×	Small businesses cannot receive or make digital payments
×	Rural populations have no mechanism to save, borrow, or insure
×	Bank infrastructure is economically unviable in low-density areas

// KEY METRICS

67% of Kenya's GDP flows through mobile money
1.4B global mobile money accounts active
\$1T+ annual transaction volume globally

// WHAT THE SYSTEM DOES

- Real-time peer-to-peer and business payment processing via USSD — no smartphone required
- Agent network management system balancing float liquidity across thousands of cash-in/cash-out points
- Fraud detection engine processing every transaction against behavioral models in under 100ms
- Regulatory compliance module auto-generating CBK audit trails and AML transaction reports
- Cross-network interoperability layer enabling transfers between competing mobile money providers
- Merchant payment gateway with QR and USSD options for small business point-of-sale integration
- Micro-savings and micro-loan engine using transaction history as the credit scoring input

// DESIGN CONSTRAINT

The system must function at full capability on a KSh 2,000 feature phone over a 2G network with 800ms latency. Any architecture decision that violates this constraint is rejected regardless of its other merits.

AI Health Diagnostics

An AI-powered clinical decision support system giving rural health workers specialist-level diagnostic capability without requiring a specialist on-site.

// PROBLEMS IT SOLVES

×	1 doctor per 5,000 people across most of Sub-Saharan Africa
×	Specialist care concentrated in urban centres — inaccessible to rural patients
×	Preventable diseases (TB, cervical cancer, retinopathy) caught too late
×	EHR data exists but is never queried for population-level insight
×	Misdiagnosis rates high where clinical experience is limited

// KEY METRICS

94% AI dermatology diagnostic accuracy vs 87% physician average
400M people with no access to specialist healthcare
40% reduction in diagnostic errors with AI-assisted review

// WHAT THE SYSTEM DOES

- Medical imaging analysis pipeline — detects anomalies in X-rays, skin scans, and retinal images using trained CNNs
- Symptom-to-differential-diagnosis engine correlating clinical inputs against validated medical knowledge bases
- Predictive risk scoring model flagging high-risk chronic disease patients before acute deterioration
- Clinical NLP module extracting structured diagnosis data from free-text physician notes
- Federated learning infrastructure — models improve across hospital networks without sharing patient data
- FHIR-compliant integration layer connecting to existing EHR and laboratory systems
- Explainability panel surfacing evidence behind every AI recommendation so clinicians can verify before acting

Every AI recommendation must display its confidence level, its evidence basis, and a clear escalation path when confidence is below threshold. The system never presents output as a final diagnosis — it is a decision support tool, not a decision-making tool.

PYTHON	TENSORFLOW	ANTHROPIC API	FHIR R4	LANGCHAIN	VECTOR DBS	DOCKER	AWS SAGEMAKER
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Precision Agriculture

A data-driven farm intelligence platform that synthesizes satellite, sensor, and market data into actionable decisions for smallholder farmers — delivered offline on basic Android devices.

// PROBLEMS IT SOLVES

- × 50–70% yield gap between actual and potential output across SSA farms
- × Farmers make planting decisions on tradition, not data — as climate shifts
- × Crop stress and disease are identified late — after damage is irreversible
- × Irrigation water wasted because soil moisture data is not available
- × Farmers sell at the wrong time due to no access to price intelligence

// KEY METRICS

70%

average yield gap on SSA smallholder farms

500M

smallholder farming families globally

25%

average water savings with precision irrigation scheduling

// WHAT THE SYSTEM DOES

- Satellite NDVI analysis pipeline detecting crop stress zones 2 weeks before symptoms are visible to the eye
- Hyper-local 14-day weather forecast engine calibrated to field-level coordinates, not regional averages
- Soil health inference model estimating NPK requirements from spectral data and input history
- Pest and disease early warning — correlates temperature, humidity, and historical outbreak data by crop type
- Commodity price intelligence showing optimal sell windows based on market trends and transport cost models
- Planting calendar generator aligned to local crop varieties, soil types, and rainfall probability curves
- Offline-first Android app with full functionality cached locally; SMS fallback for zero-data environments

// DESIGN CONSTRAINT

All recommendations are output in plain Kiswahili or the user's local language. No agronomic jargon. No charts that require interpretation. The system outputs one clear action per decision point: plant now, wait 3 days, apply 2kg/acre of X.

Digital Identity System

A privacy-first, biometric-enabled digital identity infrastructure that issues verifiable credentials citizens can use across all government and private services without surrendering their data.

// PROBLEMS IT SOLVES

×	1 billion people globally cannot legally prove who they are
×	50+ government databases hold duplicate, inconsistent citizen records
×	Service access requires physical presence at multiple offices for ID verification
×	Citizens have no control over who accesses their personal data
×	Existing ID systems exclude marginalized populations during enrollment

// KEY METRICS

1B people without legal identity worldwide
\$3T estimated economic value unlocked by functional digital ID
50+ Kenyan government databases without interoperability

// WHAT THE SYSTEM DOES

- Biometric enrollment pipeline with deduplication engine preventing duplicate or fraudulent identity registration
- Verifiable credential issuance using W3C DID standards — citizens share only the attributes each service needs
- Interoperability API allowing all government ministries to verify identity without accessing raw personal data
- Offline verification module using cryptographic proofs that work without internet connectivity
- Consent management dashboard — citizens see every entity that has accessed their data and can revoke access
- Tamper-evident audit trail logging every identity transaction permanently and immutably
- Inclusive enrollment support for people with disabilities, no fixed address, or non-standard biometrics

The system architecture must make mass surveillance structurally impossible — not just policy-prohibited. No central authority holds the full identity graph. Verification is always attribute-specific, never identity-dump. Privacy is enforced in code, not in terms of service.

NEXT.JS

NODE.JS

POSTGRESQL

W3C DID

ZERO-KNOWLEDGE PROOFS

BIOMETRIC SDK

DOCKER

AWS GOV CLOUD

Adaptive Learning Platform

An AI-driven education system that adjusts content, pacing, and difficulty in real time per student — built for low-connectivity environments and localized for Kenyan curriculum and context.

// PROBLEMS IT SOLVES

×	300 million school-age children have no access to formal education
×	Classroom instruction moves at one pace regardless of individual mastery
×	Teachers lack data to identify which students need intervention and on what topic
×	Western EdTech platforms use non-local context, reducing learning transfer
×	Low-bandwidth rural schools cannot use cloud-dependent education tools

// KEY METRICS

2× learning rate improvement with adaptive vs fixed-pace instruction
300M children globally with no school access
62% Sub-Saharan Africa mobile internet penetration — delivery channel

// WHAT THE SYSTEM DOES

- Knowledge graph mapping all CBC curriculum concepts and prerequisite dependencies across subjects and grade levels
- Bayesian knowledge tracing model estimating each student's mastery probability per concept in real time
- Content recommendation engine selecting the next best learning activity based on mastery state and engagement data
- Multi-modal content delivery — text, audio, video, and interactive simulation matched to learning style signals
- Teacher intervention dashboard surfacing class-wide knowledge gaps ranked by severity and student count affected
- AI tutoring assistant answering student questions in plain language without requiring a teacher present
- Offline-first architecture with full content library pre-loaded for schools with unreliable connectivity

All content is localized to Kenyan curriculum, names, currency, geography, and cultural references. A student in Kisii County should never encounter a math problem about subway tokens or a reading passage about snow. Localization is not translation — it is rebuilding the context from scratch.

NEXT.JS	TYPESCRIPT	ANTHROPIC API	LANGCHAIN	VECTOR DBS	MONGODB	FIREBASE	REACT NATIVE
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Urban Mobility & Logistics

A real-time supply-and-demand matching platform for urban transport and last-mile delivery — optimizing routes, reducing spoilage, and connecting drivers and goods to customers at scale.

// PROBLEMS IT SOLVES

×	Nairobi commuters average 2.5 hours daily in unoptimized traffic
×	30% of agricultural produce spoils before reaching the market
×	Delivery routes planned manually — no dynamic rerouting on traffic events
×	Cold chain breaks because temperature breaches go undetected in transit
×	Small businesses cannot afford dedicated delivery — limiting customer reach

// KEY METRICS

\$4.5T global logistics market annually
30% food lost to last-mile logistics failure in SSA
2.5hrs average daily Nairobi commute time

// WHAT THE SYSTEM DOES

- Real-time geospatial matching engine pairing available drivers with passengers or delivery requests under 3 seconds
- Dynamic route optimization recalculating paths continuously against live traffic feeds and road closure data
- Fleet telematics dashboard — vehicle utilization, fuel efficiency, driver behavior scoring, and maintenance alerts
- Cold chain IoT integration — triggers real-time alerts when cargo temperature exits safe range during transit
- Driver earnings management with surge pricing and zone-based incentive modeling to balance supply across demand
- Proof-of-delivery workflow with digital signature, GPS-stamped photo capture, and automated customer notification
- Shared delivery pooling engine — aggregates multiple small-business deliveries into single optimized routes

The matching engine must function with degraded GPS accuracy in dense urban environments and must not fail gracefully — it must fail predictably, with deterministic fallback to the last known good state. Data loss during a trip is not acceptable under any network condition.

REACT NATIVE

NODE.JS

POSTGRESQL

POSTGIS

KAFKA

REDIS

GOOGLE MAPS API

IOT MQTT

Climate Early Warning System

A multi-source climate intelligence platform that converts scientific model outputs into actionable community-level alerts — 72 hours before a flood, drought, or disease outbreak hits.

// PROBLEMS IT SOLVES

×	Climate disasters cost SSA over \$6B annually — mostly preventable with warning
×	Scientific forecasts exist but never reach the communities that need them
×	Flood inundation models are not calibrated to local terrain or river systems
×	Disease outbreak signals exist in health data but are never monitored in real time
×	Alert dissemination is slow, fragmented, and relies on broken official channels

// KEY METRICS

\$6B annual climate disaster economic losses in SSA
10× ROI of early warning investment vs disaster response cost
72hrs minimum lead time needed for effective community action

// WHAT THE SYSTEM DOES

- Multi-source ingestion pipeline — satellite imagery, ground weather stations, river gauges, and historical climate records
- Ensemble weather model combining outputs from multiple global climate models for improved regional accuracy
- Flood inundation simulator using digital elevation models and hydrological flow analysis to map at-risk zones
- Disease outbreak detection correlating environmental triggers with syndromic surveillance data from health facilities
- Multi-channel alert dissemination — SMS to individuals, radio station API integration, chief and ward officer dashboards
- Government emergency coordination module — resource pre-positioning, evacuation route mapping, and response tracking
- Post-event feedback loop — observed outcomes calibrate model parameters for improved future accuracy

An alert that reaches a government dashboard but not a community member has failed. The system measures success by confirmed community receipt and action taken — not by alert generation. The last mile of communication is treated as a core engineering problem, not a communications problem.

PYTHON

AWS

POSTGRESQL

TENSORFLOW

NETCDF

GDAL

SMS GATEWAY

MAPBOX

Government Digitalisation

An end-to-end digital service delivery platform replacing paper-based government workflows with authenticated, auditable, mobile-first citizen journeys — across licensing, permits, compliance, and public records.

// PROBLEMS IT SOLVES

×	Business registration takes weeks due to manual, multi-office paper workflows
×	Citizens must physically appear at offices that operate limited hours
×	No real-time visibility into application status — citizens call, wait, and repeat
×	Corruption enabled by cash handling and manual approval discretion
×	Government ministries cannot share verified citizen data — duplication everywhere

// KEY METRICS

82%

cost reduction in digitized vs manual government service delivery

60%

corruption reduction reported with full digital service delivery

40hrs

average annual time SMEs spend on government compliance paperwork

// WHAT THE SYSTEM DOES

- Unified citizen portal with single sign-on valid across all participating government departments and agencies
- Configurable approval workflow engine supporting multi-stage, multi-officer processes with SLA enforcement
- Document management module with e-signature, version history, and tamper-evident storage
- Integrated payment layer — M-Pesa, bank transfer, and card payment with automatic fee calculation and receipt generation
- Citizen service tracker — real-time application status with SMS notifications at each stage transition
- Administration analytics dashboard — bottleneck identification, officer performance metrics, and SLA breach alerts
- Verified data API — allows third-party services (banks, insurers) to confirm citizen attributes without re-collecting data

MIKESTH3TIC.DEV is currently mapping and designing the digitalisation architecture for the State Department of Youth Affairs, Creative Economy and Sports — covering KECOBO, DITD, and NYC service workflows for automation and citizen-facing integration.

NEXT.JS

TYPESCRIPT

NODE.JS

POSTGRESQL

M-PESA API

FIREBASE AUTH

DOCKER

AWS

ORACLE — Investment Intelligence

An AI-powered NSE investment intelligence platform giving Kenyan retail investors institutional-grade analysis — fundamental data, technical signals, sentiment monitoring, and AI-generated briefings — in a single consumer-grade interface.

// PROBLEMS IT SOLVES

×	Retail investors have no structured access to NSE fundamental data
×	Annual reports and statutory filings are unstructured PDFs nobody can query
×	No platform surfaces NSE technical analysis in plain, actionable language
×	Retail investors make decisions on tips and rumour — not evidence
×	Under 5% of Kenyan adults hold investment accounts despite \$14B market cap

// KEY METRICS

65+ companies listed on the NSE
\$14B NSE total market capitalization
<5% Kenyan adults with active investment accounts

// WHAT THE SYSTEM DOES

- Automated fundamental analysis — parses NSE filings, annual reports, and earnings releases into structured financial metrics
- Technical analysis engine computing 40+ indicators (RSI, MACD, Bollinger, moving averages) with plain-language interpretation
- AI sentiment pipeline monitoring financial news, company announcements, and social signals per listed company
- Portfolio optimization tool running mean-variance analysis against user-defined risk tolerance and investment horizon
- Earnings calendar with AI-generated pre-earnings briefings synthesizing historical performance and analyst expectations
- Sector comparison engine benchmarking any company against direct competitors and sector-wide averages
- Watchlist alert system — price movement, unusual volume, regulatory announcements, and earnings date proximity

ORACLE does not give buy or sell recommendations. It gives the evidence. A user who understands why a metric matters is a better investor than one who followed a signal they didn't understand. Every data point surfaces with an explanation of what it means, not just what it says.

NEXT.JS

PYTHON

ANTHROPIC API

LANGCHAIN

VECTOR DBS

POSTGRESQL

NSE DATA API

TAILWIND CSS

Supply Chain & Market Linkage

A verified B2B marketplace connecting producers directly to buyers — with digital contracts, payment escrow, quality verification, and logistics coordination built in to eliminate the intermediaries that extract margin without creating value.

// PROBLEMS IT SOLVES

- × 40–60% of farm-gate value captured by intermediaries who add no value
- × Buyers cannot verify supplier quality, capacity, or reliability before transacting
- × No digital contract layer — disputes resolved informally with no recourse
- × 30% post-harvest losses from poor logistics coordination and delayed sale
- × Small producers cannot access export markets they cannot reach or navigate alone

// KEY METRICS

40%

farm-gate value lost to middlemen in Kenya

30%

post-harvest losses from supply chain inefficiency

\$1.2B

Kenya agricultural export value at risk from logistics failure

// WHAT THE SYSTEM DOES

- Verified supplier and buyer marketplace — KYB-compliant profiles, transaction history, and quality ratings per entity
- Real-time price discovery engine aggregating live offers to surface market-clearing prices by commodity and grade
- Digital contract generation with e-signature, delivery terms, and M-Pesa escrow tied to confirmed delivery
- Quality assurance workflow — photo capture at source, third-party certification integration, and structured dispute resolution
- Logistics matching — connects transacting parties to vetted transporters with real-time delivery tracking
- Demand forecasting module — shows suppliers when and where demand is rising so they can plan production accordingly
- Export readiness engine — identifies which producers meet standards for specific export markets and connects them to buyers

Agri Value Connect, built by MIKESTH3TIC.DEV, operationalizes this architecture for Kenya's agricultural sector — connecting smallholder farmers directly to institutional buyers and processing the full transaction, quality check, and payment workflow on a single platform.

REACT

NODE.JS

POSTGRESQL

M-PESA API

STRIPE

TWILIO

DOCKER

AWS

10 SYSTEMS. BUILT TO SOLVE REAL PROBLEMS.

DISRUPT · AUTOMATE · DOMINATE

SYS.01 Mobile Money Infrastructure Banking 2.5B unbanked adults	SYS.02 AI Health Diagnostics Specialist care where no specialists are
SYS.03 Precision Agriculture Closing the 70% SSA yield gap	SYS.04 Digital Identity System Legal existence for 1B invisible people
SYS.05 Adaptive Learning Platform Quality education at scale, offline-first	SYS.06 Urban Mobility & Logistics Ending the 2.5hr Nairobi commute
SYS.07 Climate Early Warning 72hrs before the flood hits	SYS.08 Government Digitalisation From 3 weeks to 20 minutes
SYS.09 ORACLE — Investment Intel NSE intelligence for every Kenyan	SYS.10 Supply Chain & Linkage Returning 40% margin to producers

